Mikhail Razakov

Software Engineer — Systems & Architecture

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I build correctness-sensitive software across compiler infrastructure, offline web/Android products, and backend recovery workflows. In UniversalToolchain, botay!, and VpnMediator, I make state and decision boundaries explicit, then verify the result rather than relying on the happy path.

2022: Compilers / VMs · 2023: CIL + native x86-64 · 2024–2025: AST / IR / backends → toolchain · 2026 →: LLVM + program analysis

Compiler infrastructure

UniversalToolchain: immutable, independently checked semantic plans before execution.

End-to-end product

botay!: browser/Android clients, offline operation replay and validated planning witnesses.

Reliability

VpnMediator: durable payment evidence, idempotent access operations and quarantine.

EXPERIENCE

July — August 2026

MCST — Compiler Engineering Intern

- Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety.
- Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases.

PROJECTS

UniversalToolchain / Wist2

Built one semantic planning boundary: feature dependencies, conflicts and typed backend routes become an immutable LanguagePlan before runtime execution.

A separate verifier checks exact package identity and routes; architecture gates and a 1,324-test manifest passed at the inspected 9 Oct 2026 revision.

botay! / student-execution-os

Built web/Android task planning with separate owners for external evidence, canonical user state, derived schedules and LLM suggestions.

Offline commands replay idempotently; feasibility requires independently checked witnesses. Signed Android release 0.8.3 passed its workflow on 10 Oct 2026.

VpnMediator

Persisted payment-provider evidence before subscription/access changes; idempotent entitlements and leased reconciliation handle retries and uncertain outcomes.

Ambiguous remote/local divergence is quarantined for review; recovery includes backup/restore and rollback safeguards. Deployment scale is not asserted.

SKILLS

Languages / platforms

C++23, C#/.NET, Python, Rust, Java, JavaScript, SQL, Linux

Architecture

ownership boundaries, typed contracts, immutable planning, canonical vs derived state

Product / backend

offline-first sync, Android, SQLite/PostgreSQL, payment workflows, migrations

Correctness

independent verifiers, differential tests, idempotency, reconciliation and recovery

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

HSE Faculty of Computer Science — open-source projects competition

Selected as one of the winners with UniversalToolchain / Wist2.

LangDev'26 — Build the Language, Then Make the Abstractions Disappear

Delivered the UniversalToolchain architecture talk at LangDev on 9 October 2026.

Moscow / remote